

Question description

Question No:7

You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i , output 1 as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

$$1 \leq N \leq 3000$$

$$0 \leq A[i] \leq 10^{18}$$

Input :

The first line contains a single integer N denoting the size of array A . Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$.

Output :

Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or 1 , if $G(F(i))$ does not exist.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

8
3 7 1 7 8 4 5 2

EXPECTED OUTPUT

1 4 4 4 -1 2 -1 -1

Test Case 2

INPUT (STDIN)

8
31 71 11 17 81 14 51 21

EXPECTED OUTPUT

11 14 14 14 -1 21 -1 -1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(a[j]>a[i])

Test Case 2

KEYWORD

for(i=0;i<n;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

30

Code :

```
1  #include <stdio.h>
2  #define MAX 3000
3  int main()
4  {
5      int N;
6      int A[MAX];
7      int F[MAX];
8      int G[MAX];
9
10     scanf("%d", &N);
11
12     for (int i = 0; i < N; i++)
13     {
14         scanf("%d", &A[i]);
15     }
16
17     /* Find F(i): first element greater than A[i] */
18     for (int i = 0; i < N; i++)
19     {
20         F[i] = -1;
21
22         for (int j = i + 1; j < N; j++)
23         {
24             if (A[j] > A[i])
25             {
26                 F[i] = j;
27                 break;
28             }
29         }
30     }
31
32     /* Find G(i): first element smaller than A[i] */
33     for (int i = 0; i < N; i++)
34     {
35         G[i] = -1;
36
37         for (int j = i + 1; j < N; j++)
```

```
38 {
39     if (A[j] < A[i])
40     {
41         G[i] = j;
42         break;
43     }
44 }
45 }
46
47 /* Find A[G(F(i))] */
48 for (int i = 0; i < N; i++)
49 {
50     if (F[i] == -1)
51     {
52         printf("-1");
53     }
54     else if (G[F[i]] == -1)
55     {
56         printf("-1");
57     }
58     else
59     {
60         printf("%d", A[G[F[i]]]);
61     }
62
63     if (i < N - 1)
64     {
65         printf(" ");
66     }
67 }
68
69 return 0;
70 }
```

Output:

Test Case-1

```
8
3 7 1 7 8 4 5 2
1 4 4 4 -1 2 -1 -1
-----
Process exited after 12.43 seconds with return value 0
Press any key to continue . . . |
```

Test Case-2

```
8
31 71 11 17 81 14 51 21
11 14 14 14 -1 21 -1 -1
-----
Process exited after 3.885 seconds with return value 0
Press any key to continue . . . |
```

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